Preliminary validation of portable 3D-force instrumented insoles for cycling power output estimation: a proof-of-concept case study

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Introduction

Cycling is widely utilized in neurological rehabilitation due to its constrained kinematics and low fall risk compared to walking [1]. While characterizing pedaling metrics like power efficiency and symmetry can provide crucial insights into a patient's clinical status [2], real-time power estimation using wearable, portable sensors remains unfeasible, limiting compatibility with different ergometers. To address this gap, this study evaluates the feasibility of using 3D-force instrumented insoles for clinical cycling power estimation, presenting a proof-of-concept evaluation with a single healthy participant.

Methods

Protocol

A female healthy participant performed a 45-second pedaling trial at 60 rpm and resistance level 6 on a stationary bicycle equipped with instrumented X-power pedals (SRM, Germany). The participant's dominant (right) foot was instrumented with Shokac shoe insoles designed by Nakai et al. [3] featuring three sensors located at the thumb, fifth metatarsal, and heel giving force data in three directions.

Data analysis

The foot-to-crank angle (θ) was measured using two inertial measurement units (X-io Technologies, UK) on a different trial. The IMUs were in the crank center of rotation and external part of the foot and the relative angle in the sagittal plane was calculated using quaternions. Consequently, the insole's axes were mapped directly to the pedal, using the formulas:

$$F_{rad} = F_z * \cos(\theta - 90), F_{tang} = F_z * \sin(\theta - 90)$$

Where $\boldsymbol{F_z}$ was the sum of the perpendicular component of the three sensors. The mean and standard deviation of the 30 central cycles were calculated to remove start/end deviations. Pearson's correlation (r), coefficient of determination (R^2) and Root means squared error (RMSE) were computed to evaluate signal morphology and temporal alignment. To assess continuous local differences, Statistical Parametric Mapping (SPM) for paired designs (*spm1d.ttest_paired*) was performed. Each cycle was min-max normalized (0-1) to isolate waveform shape from absolute magnitude differences. Testing the null hypothesis ($\boldsymbol{H_0}: \boldsymbol{\mu_{pedal}(\theta) = \mu_{insole}(\theta)}$) versus the alternative ($\boldsymbol{H_1}: \boldsymbol{\mu_{pedal}(\theta) \neq \mu_{insole}(\theta)}$), Random Field Theory (RFT) was used to determine the critical thresholds (α=0.05).

Results and discussion

Figure 1 presents the mean tangential and radial forces over a crank cycle (0-360º) comparing the insoles against the pedals. A strong temporal and phase-domain correlation and variance sharing were observed for both the tangential ($r = 0.971, R^2 = 0.943,\ RMSE = 0.085$) and radial force ($r = 0.977, R^2 = 0.955,\ RMSE = 0.104$). Point-by-point SPM analysis revealed statistically significant magnitude differences, for the tangential force, significant clusters appeared around ~30–50° and ~125–145°; whereas for the radial force, significance was observed between ~305–360° (Figure 2). Direct amplitude comparison was restricted possibly by structural contact surface area differences between the insole and the pedal interface.

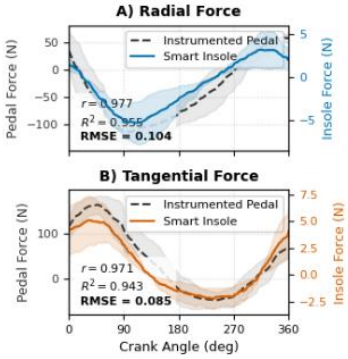


Figure 1. Radial (A) and Tangential (B) force comparison between insole and pedal with Pearson's and determination correlation.

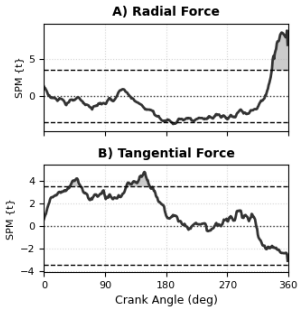


Figure 2. Radial (A) and Tangential (B) force comparison using SPM along the pedalling cycle.

Conclusions

This proof-of-concept study demonstrates the feasibility of using 3D-force instrumented insoles to estimate effective tangential and radial forces, showing high morphological agreement with gold-standard instrumented pedals. This portable solution offers universal compatibility with ergometers for accessible rehabilitation and real-time monitoring. However, this one subject study can only be taken as an example of the potential of this technology and future work should focus on increasing sample size.

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References

1. P. Kajganic, et al., *Sensors*, vol. 23, no. 7, Apr. 2023.
2. J. Soulard, et al., *Ann Phys Rehabil Med*, vol. 68, no. 5, p. 101955, Jun. 2025.
3. A. Nakai, et al., *The Proceedings of the Symposium on sports and human dynamics*, vol. 2022, no. 0, p. B-5-4, 2022.